

VISION + SENSORS

DINOv2 + PATCHCORE

RISK SCORING

4-6 WEEK PILOT

Industrial Predictive Quality MVP

Anticipate defects, failures and process deviations by combining industrial vision, sensors and process data.

A fast pilot to turn plant history into risk rankings, defect heatmaps and actionable inspection decisions.

[View product](#)

[Pilot plan](#)

The business problem

Plants already generate data; the challenge is converting it into decisions before the cost appears.

Unplanned downtime

Operational risk, delays, scrap and reactive maintenance.

Late visual defects

Manual inspection is expensive and defects are detected too late.

Underused signals

SCADA, PLC, test logs, cameras and maintenance systems are usually analysed separately.

Too many alarms

Engineering teams need prioritisation, not more noise.

What we sell as an MVP

A modular system to prioritise inspections, anticipate machine failure risk and turn industrial data into actionable decisions.

Visual Quality Foundation

Pretrained DINOv2, PatchCore-lite, PaDiM-lite, heatmaps and image/part ranking.

Machine Failure Risk Scoring

Physical sensors, process variables, engineered features, calibrated models and top 5-10% alerts.

Process Intelligence

Metadata, setpoints, recipes, test logs and operational context integrated into one score.

Predictive Quality World Model

Expected-process-evolution model to detect

early deviations from sequences and setpoints.

Three products in one pilot

The customer receives value across different data realities: cameras, sensors, SCADA, test logs or maintenance history.

Visual defect detection

Part/image heatmaps and inspection ranking for surfaces, PCB, packaging, products or video.

Machine failure risk

Failure-soon prediction using sensors, cycle, process data and operational lead time for maintenance.

Process deviation alerts

Expected vs observed process evolution to warn before scrap, downtime or quality loss.

Models integrated in the MVP

DINOv2 / ResNet

dense visual features

PatchCore / PaDiM

normal memory + heatmaps

Engineered sensor features

RMS, energy, frequency, kurtosis

Latent future features

optional learned signal

World-model surprise

expected temporal deviation

Hierarchical ranking

patch → part → cycle → batch → line

Pilot architecture

1

Plant data

Cameras, sensors, SCADA, test logs, maintenance records



2

Useful signal

Dense vision, visual memory, physical sensors



3

Models

Risk scoring, anomalies and expected deviation



4

Alerts

Top 10%, heatmaps, part/cycle/batch/line



5

Decision

Inspect, adjust and integrate

Current results that support selling a pilot

Internal validation on public industrial data: strong visual inspection signal and useful sensor signal when cycle proxies are not enough.

0.60

Pixel baseline

AUPRC · Simple visual control

1.00

DINOv2 + PatchCore

AUPRC · Quick run MVTec bottle

0.999

ResNet18 + PatchCore

AUPRC · Quick run MVTec
bottle

0.558

Sensor engineered

AUPRC · No-cycle CNC

0.550

**Future learned
signal**

AUPRC · No-cycle CNC

0.763

**Hardness/material
sensors**

AUPRC · Held-out hardness

0.720

**Sensors + learned
signal**

AUPRC · Held-out hardness

+0.035

**JEPA delta over
engineered**

AUPRC · Best hardness case

Business reading

When cycle counts or threshold checks are not enough, physical and visual signals can add actionable information. The pilot validates whether this is true in the client's own data.

Specific module: machine failure-soon risk

The sensor module is sold as risk scoring: physical signals, process context and prioritised alerts.

0.701

**Operational
metadata+ future**

0.558

**No-cycle sensor
engineered**

0.550

**No-cycle predicted
future**

signal

AUPRC, $h=3/K=10$

AUPRC without cycle proxies

AUPRC future latent signal

0.763

Held-out hardness sensor engineered

AUPRC material/hardness
shift

Product and services

Visual pilot

Surface quality, PCB,
packaging, product and
industrial video with
heatmaps.

Machine failure pilot

Risk scoring with vibration,
currents, telemetry, SCADA,
cycle, process data and
lead time.

Data audit

Which signals work, which
variables are redundant
and what can be
automated.

Executive demo

HTML/PDF, tables,
inspection rankings and
integration roadmap.

Priority accounts in Galicia

Public outreach contacts. This does not imply a relationship, interest or partnership.

1 • EMAIL

FINSA

Madera / tableros

Surface quality, process sensors and predictive maintenance in production lines.

finsa@finsa.es · +34 981 050 000

2 • EMAIL

Televés

Electrónica

Test logs, PCB inspection, production anomaly detection and quality control.

televes@televes.com · 981 52 22 00

3 • EMAIL

Norvento

Energía

SCADA, vibration, early alerts and predictive maintenance for energy assets.

norvento@norvento.com · +34 982 227889

4 • EMAIL

Marine Instruments

Telemetría marina

Marine telemetry, sensors and early anomaly detection in equipment fleets.

marineinstruments@marineinstruments.es · 986366360

5 • EMAIL

Alén Space

Aeroespacial

Satellite telemetry, sensor alerts and operational anomaly ranking.

info@alen.space

6 • WEB FORM / CALL

Cinfo

Vídeo / visión

Video analytics, visual anomaly detection, events and industrial vision pilots.

Formulario web · +34 881 896 975

4-6 week pilot

1. Ingestion

Historical data, images, sensors, logs and quality/failure events.

2. Baselines

Simple rules, classical models and strong visual baselines.

3. Modelling

Risk scoring, anomaly detection, heatmaps and top 10% ranking.

4. Decision

Investment report, local demo and integration plan.

Pilot deliverables

- ✓ Benchmark with real metrics by category or asset.
- ✓ Visual heatmaps and top inspection ranking.
- ✓ Risk score by part, cycle, batch or line when data exists.
- ✓ Executive and technical report with recommendations.
- ✓ Local HTML demo and PDF for management.

Clear scope for confident selling

Next step: 15-minute technical call and selection of one pilot line/use case.

Clear scope for confident selling

What we deliver

- ✓ Validation on the client's real data.
- ✓ Comparison against simple and strong baselines.
- ✓ Useful metrics: AUPRC, Precision@10, false alarms and lead time.
- ✓ Modular product: vision, sensors, process and alerts.

Conditions for deployment

- ✓ Enough labelled data or historical events.
- ✓ Review with plant experts.
- ✓ Pilot before operational integration.
- ✓ Monitoring and maintenance if it moves to production.

industrial_jepa_mvp project · Commercial industrial AI MVP